

Ch 2.2

Machine Learning Overview

Supervised vs. Unsupervised Learning

Supervised learning



Learns from being given “right answers”

- Supervised learning uses examples with known answers.
- Each example pairs input X with output Y .
- The model learns the relationship between them.
- This lets it predict Y for a new X , as shown in the graph.

Input (X)		Output (Y)	Application
email	→	spam? (0/1)	spam filtering
audio	→	text transcripts	speech recognition
English	→	Spanish	machine translation
ad, user info	→	click? (0/1)	online advertising
image, radar info	→	position of other cars	self-driving car
image of phone	→	defect? (0/1)	visual inspection

Two groups of supervised learning algorithms

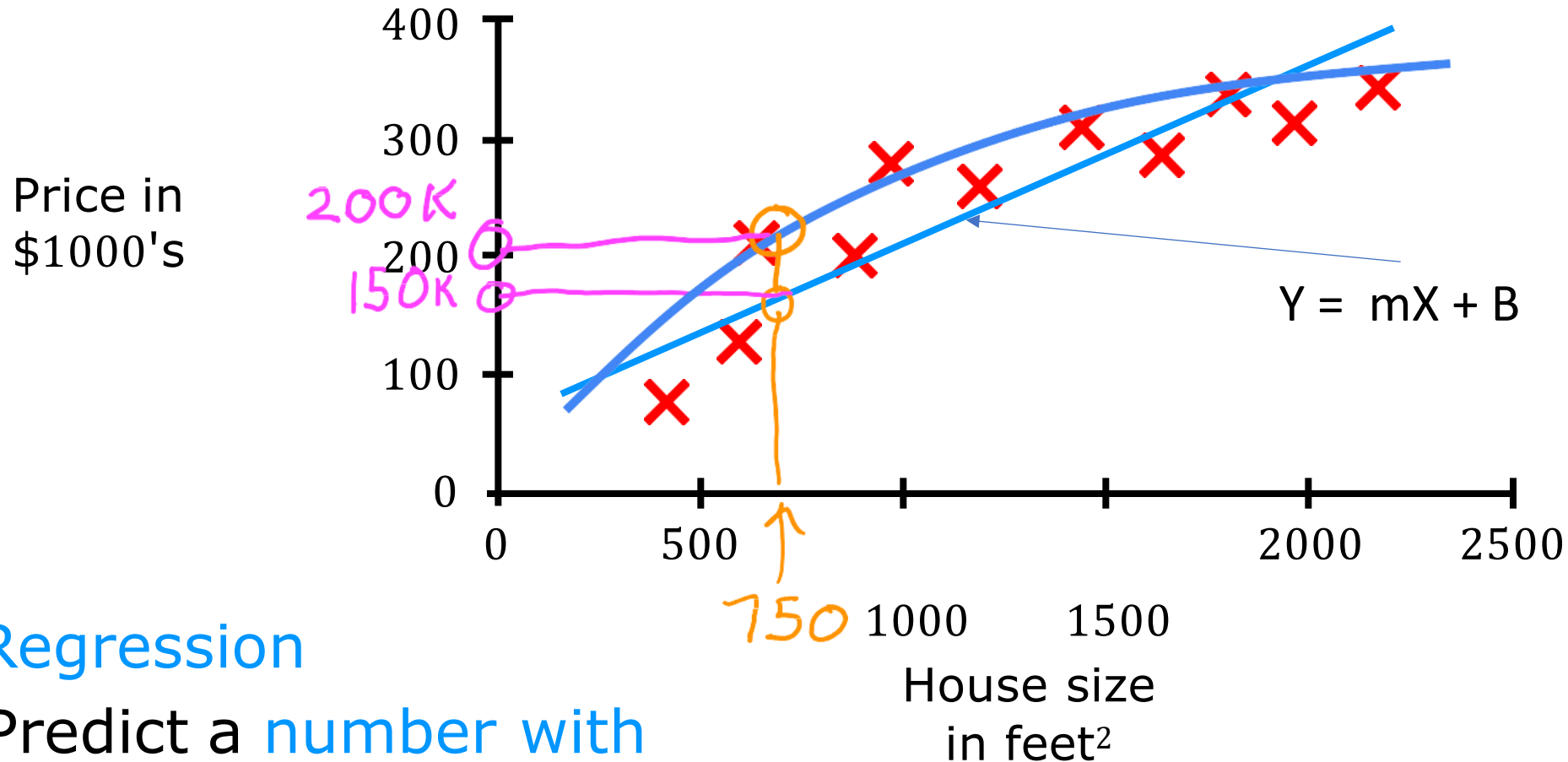
1- Regression:

- Objective: Predicts a continuous quantity or value.
- Example: Predicting house prices, stock prices, temperature forecasting.

2- Classification:

- Objective: Predicts the category or class that an input belongs to.
- Example: Email spam detection, image recognition (identifying objects in images), sentiment analysis.

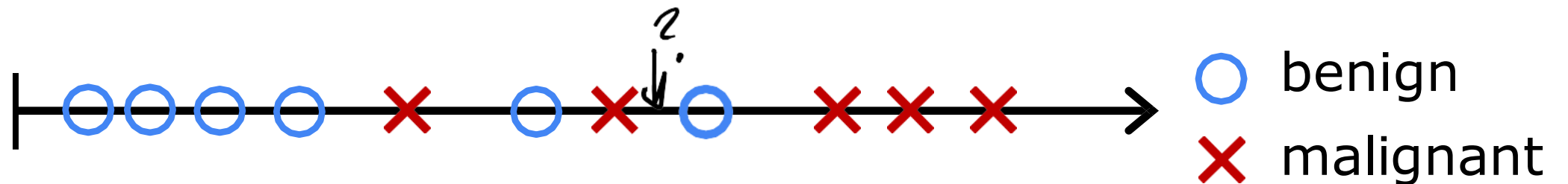
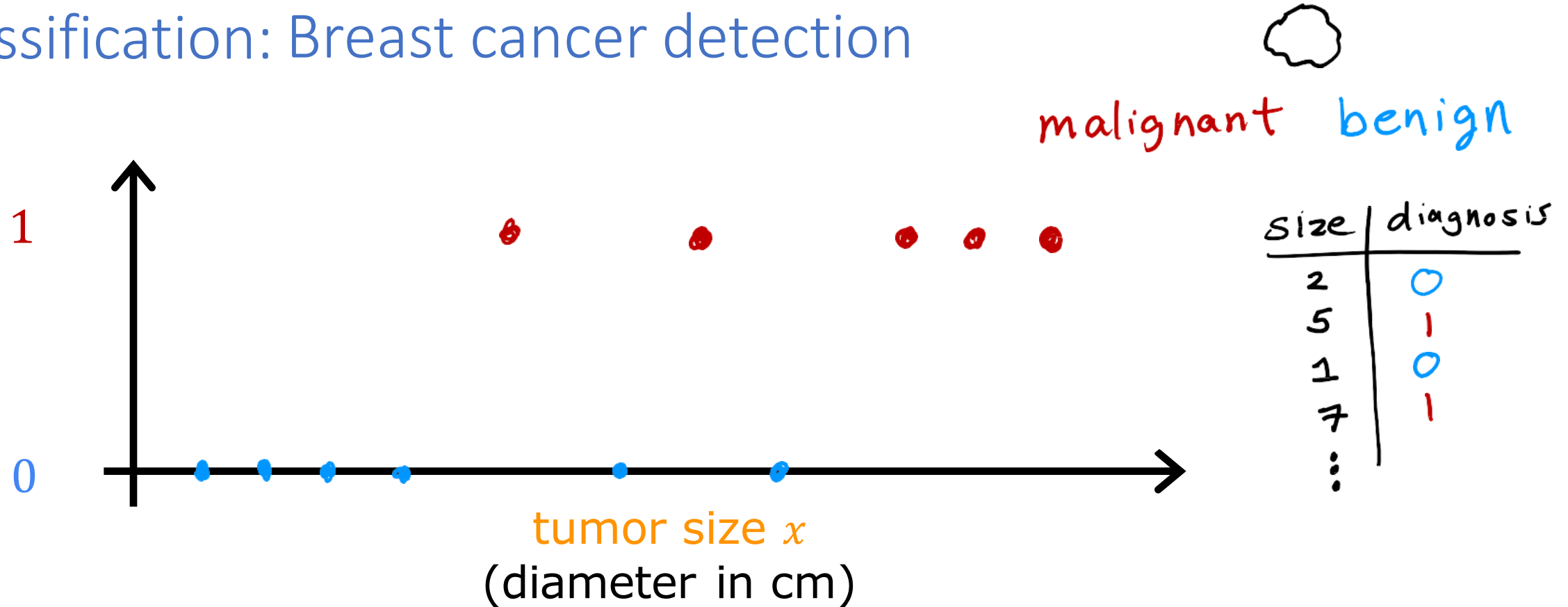
Regression: Housing price prediction



Regression

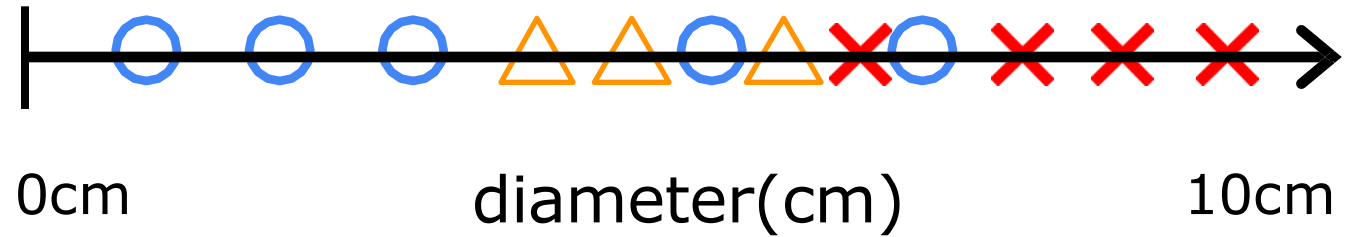
Predict a **number with infinitely** many possible outputs

Classification: Breast cancer detection



Classification: Breast cancer detection

- benign
- ✗ malignant *type1*
- △ malignant type 2



class category

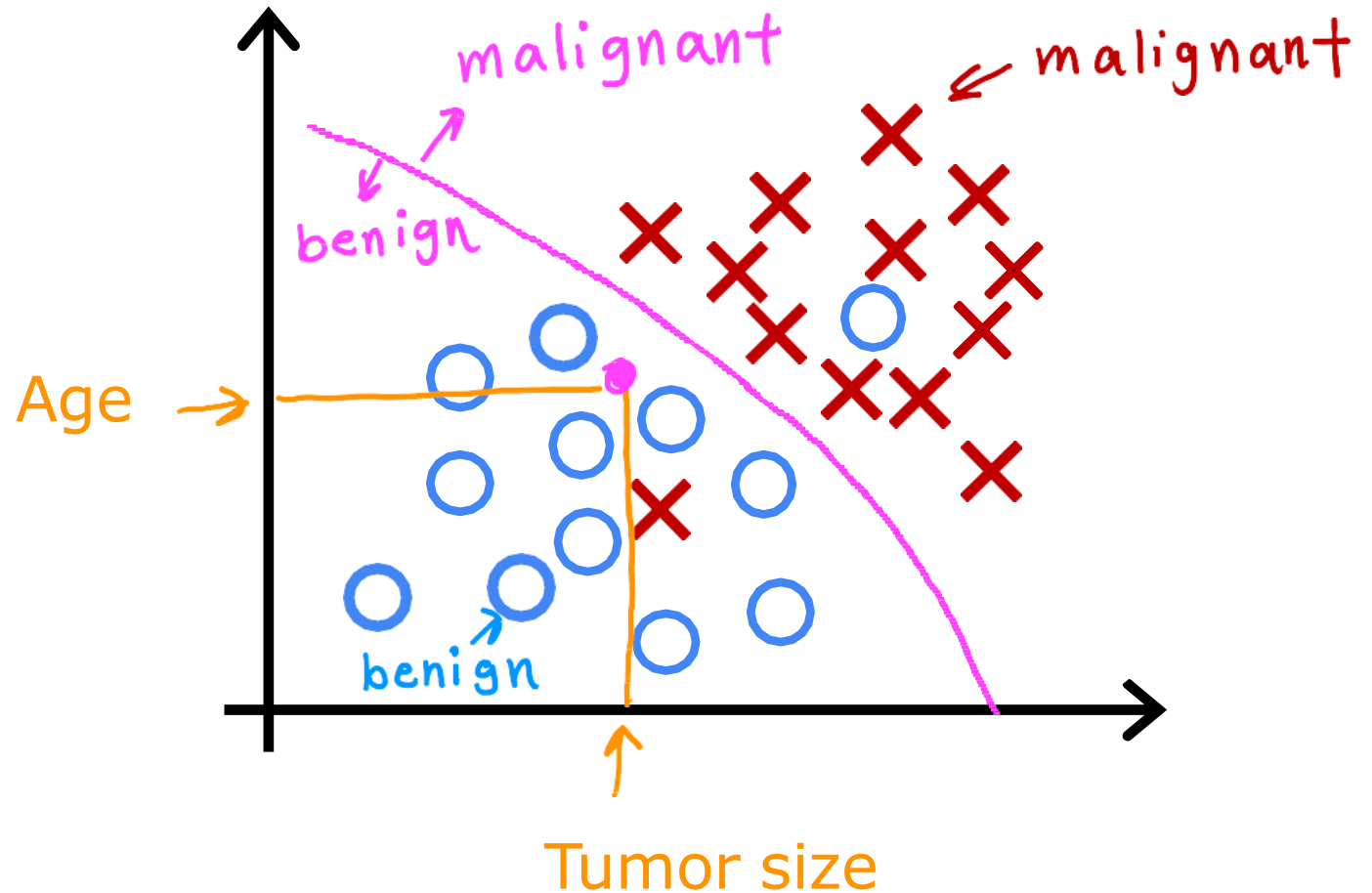
Classification

predict *categories* *cat dog* benign malignant 0, 1, 2

small number of possible outputs

Two or more inputs

For example, your learning algorithm might predict multiple types of cancer diagnoses if the tumour is malignant

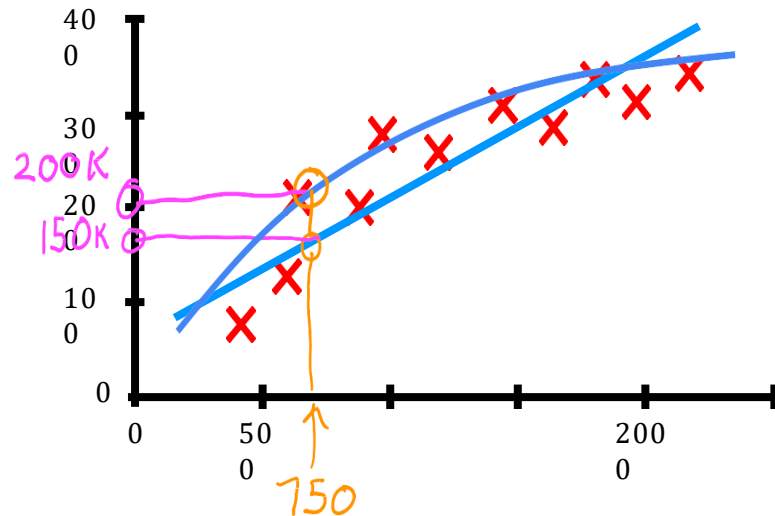


Regression vs. Classification

Learns from being given “right answers”

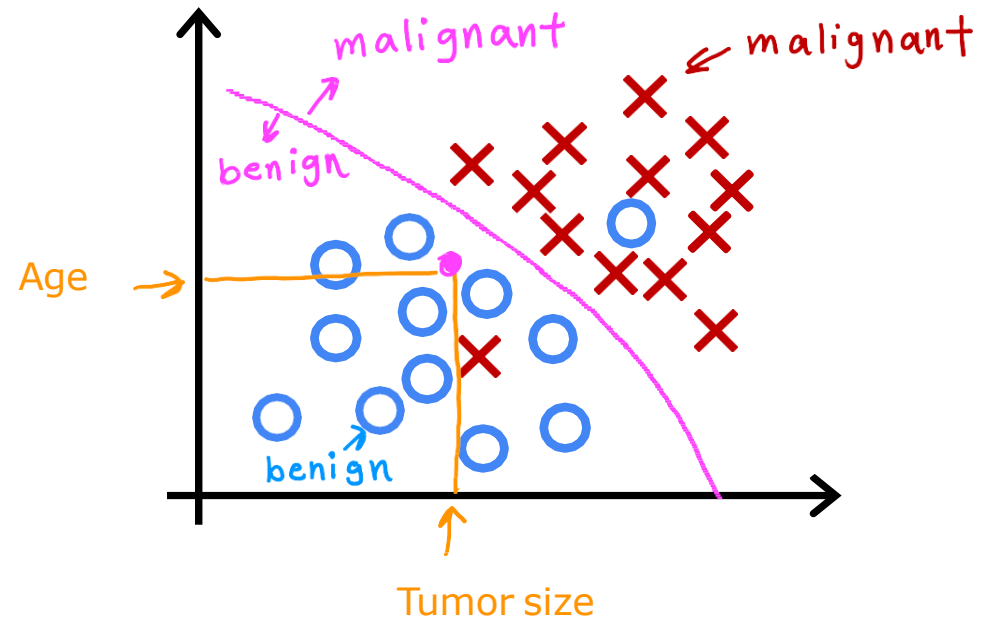
Regression

Predict a **number infinitely**
many possible outputs



Classification

predict **categories**
small number of possible outputs



Machine Learning Overview

Unsupervised Learning Part 1

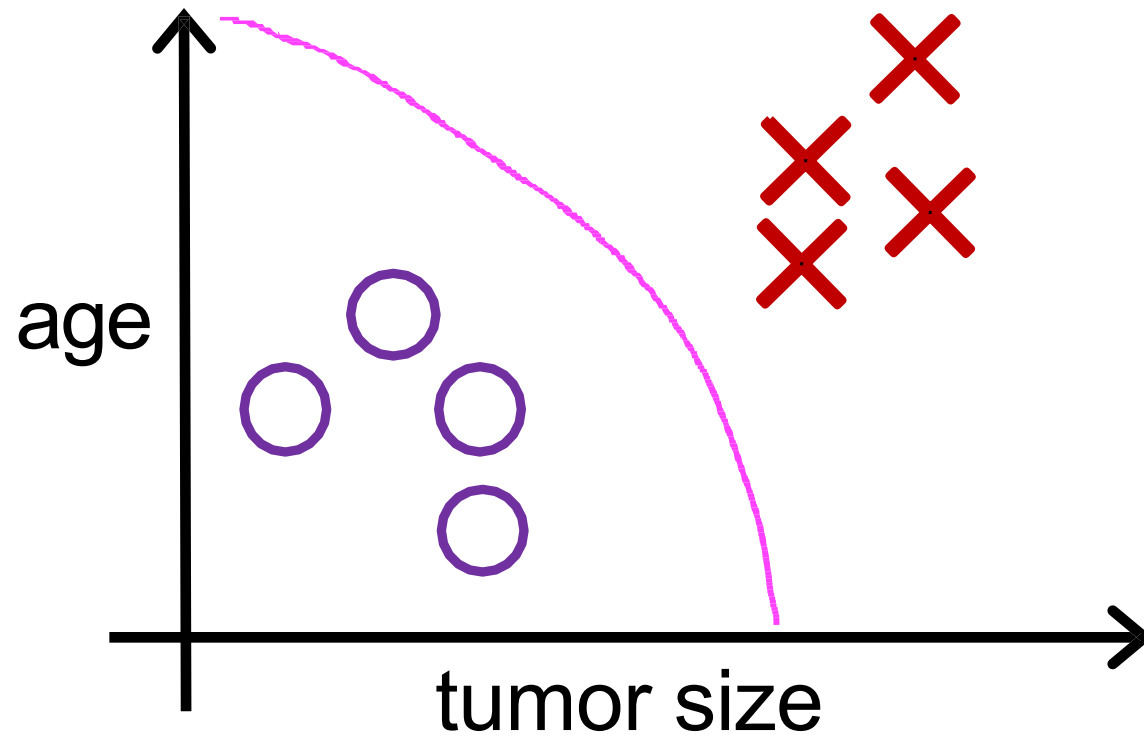
Previous: Supervised learning

In supervised learning, we focus on input-output mappings, where each example is associated with a specific output label (Y). For example, classifying tumors as benign or malignant

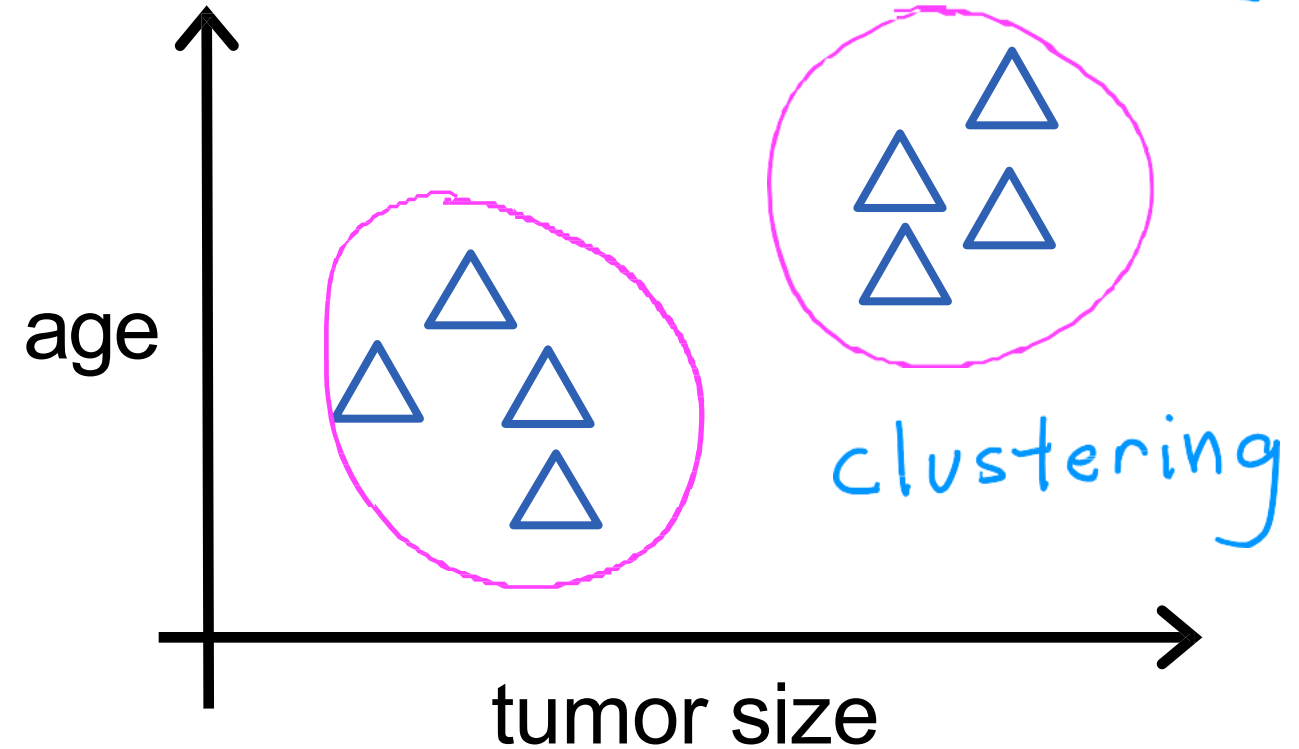
Now: UnSupervised learning

In contrast, unsupervised learning deals with data that lacks any output labels (Y). Consider a dataset with information on patients, tumor size, and age, but no information about tumor malignancy

Supervised learning
Learn from data **labeled**
with the “**right answers**”



Unsupervised learning
Find something interesting
in **unlabeled** data.



Clustering: Google news

→

Giant panda gives birth to rare twin cubs at Japan's oldest zoo

USA TODAY · 6 hours ago

• Giant panda gives birth to twin cubs at Japan's oldest zoo

CBS News · 7 hours ago

• Giant panda gives birth to twin cubs at Tokyo's Ueno Zoo

WHBL News · 16 hours ago

• A Joyful Surprise at Japan's Oldest Zoo: The Birth of Twin Pandas

The New York Times · 1 hour ago

• Twin Panda Cubs Born at Tokyo's Ueno Zoo

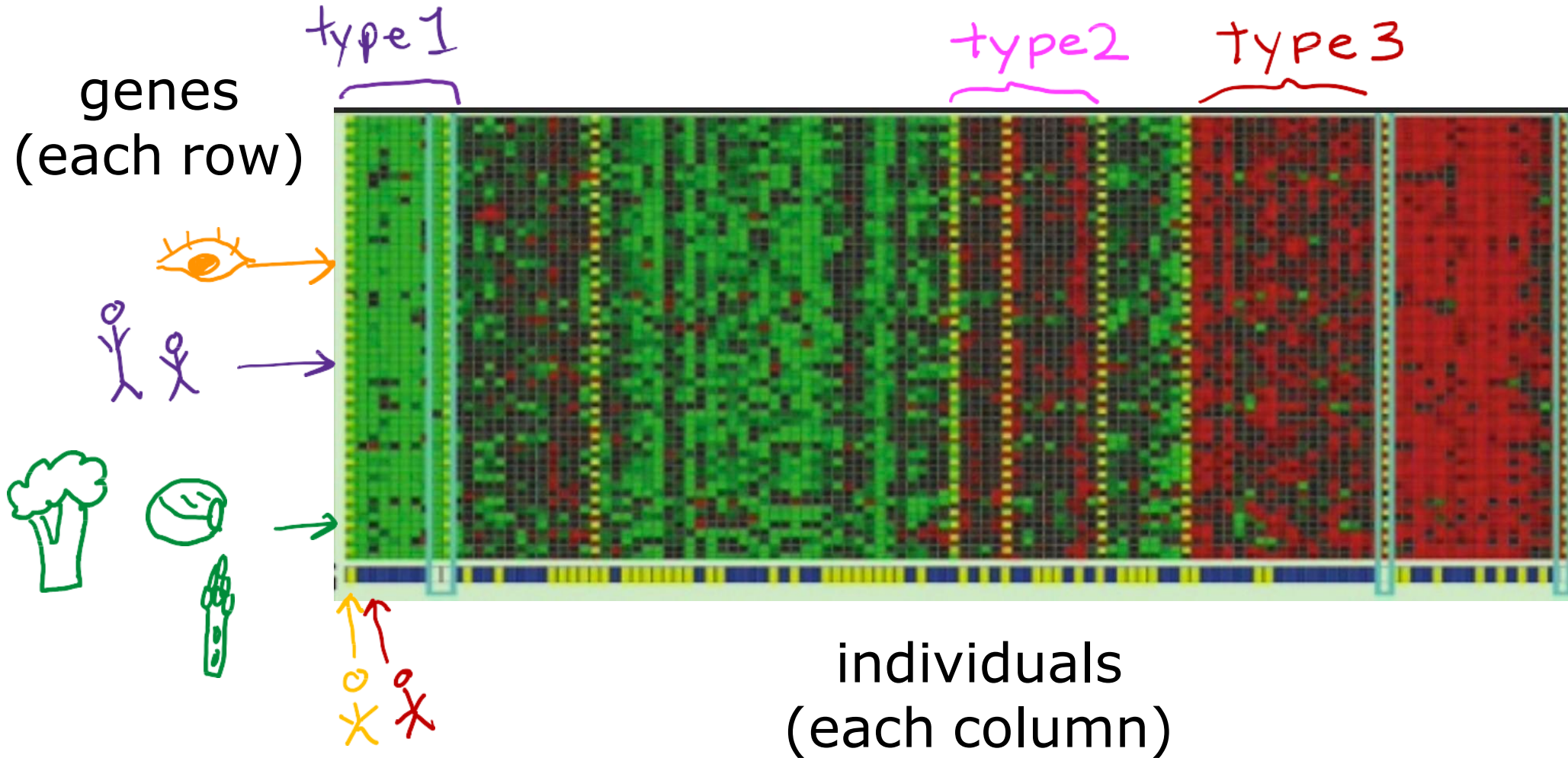
PEOPLE · 6 hours ago

A small, close-up photograph of a giant panda cub, showing its black and white fur and its face.

 View Full Coverage

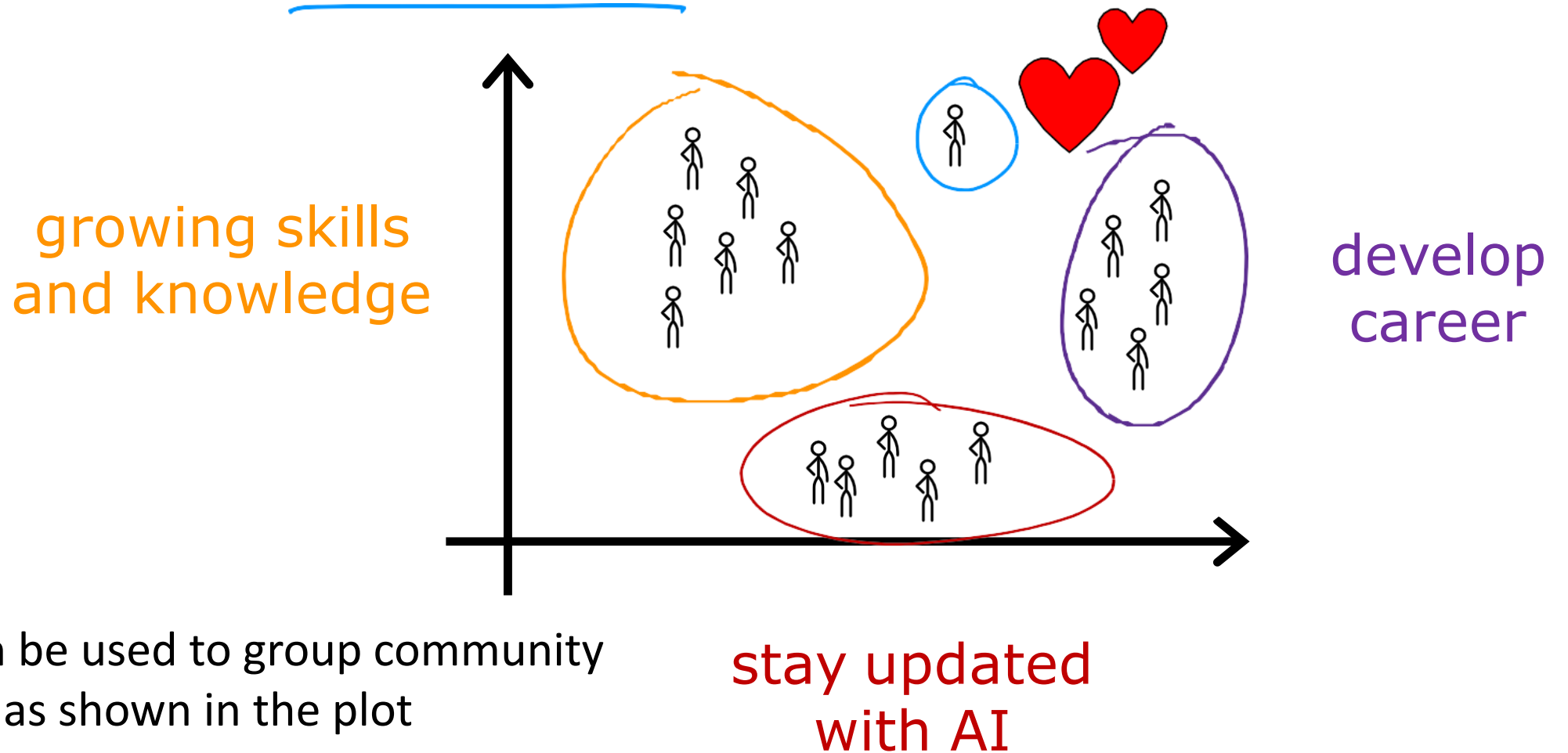
↑

Clustering: DNA microarray



Clustering: Grouping customers

Businesses can group customers into segments



Clustering can be used to group community inot clusters, as shown in the plot

Machine Learning Overview

Unsupervised Learning Part 2

Unsupervised learning

Data only comes with inputs x , but not output labels y .
Algorithm has to find **structure** in the data.

Clustering

Group similar data points together.

Dimensionality reduction

Compress data using fewer numbers.

Anomaly detection

Find unusual data points.

Question

Of the following examples, which would you address using an **unsupervised** learning algorithm?

- ☐ Given email labeled as spam/not spam, learn a spam filter.
- ☐ Given a set of news articles found on the web, group them into sets of articles about the same story.
- ☐ Given a database of customer data, automatically discover market segments and group customers into different market segments.
- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

Machine Learning Overview

Jupyter Notebooks

Jupyter Notebook - Your Interactive Learning Environment

- Jupyter Notebook is an essential tool for interactive learning in machine learning and data science.

What is Jupyter Notebook?

- Jupyter Notebook is an open-source web application that allows you to create and share documents that contain live code, equations, visualizations, and narrative text.

Why Use Jupyter Notebook?

- Interactive Learning: Jupyter Notebook allows you to experiment with code, explore data, and visualize results interactively.
- Ideal for Experiments: You can run code one line at a time, making it perfect for trying out different machine learning algorithms and concepts.
- Document Your Work: Combine code, text, and visualizations in a single document, making it easy to share and understand your work.